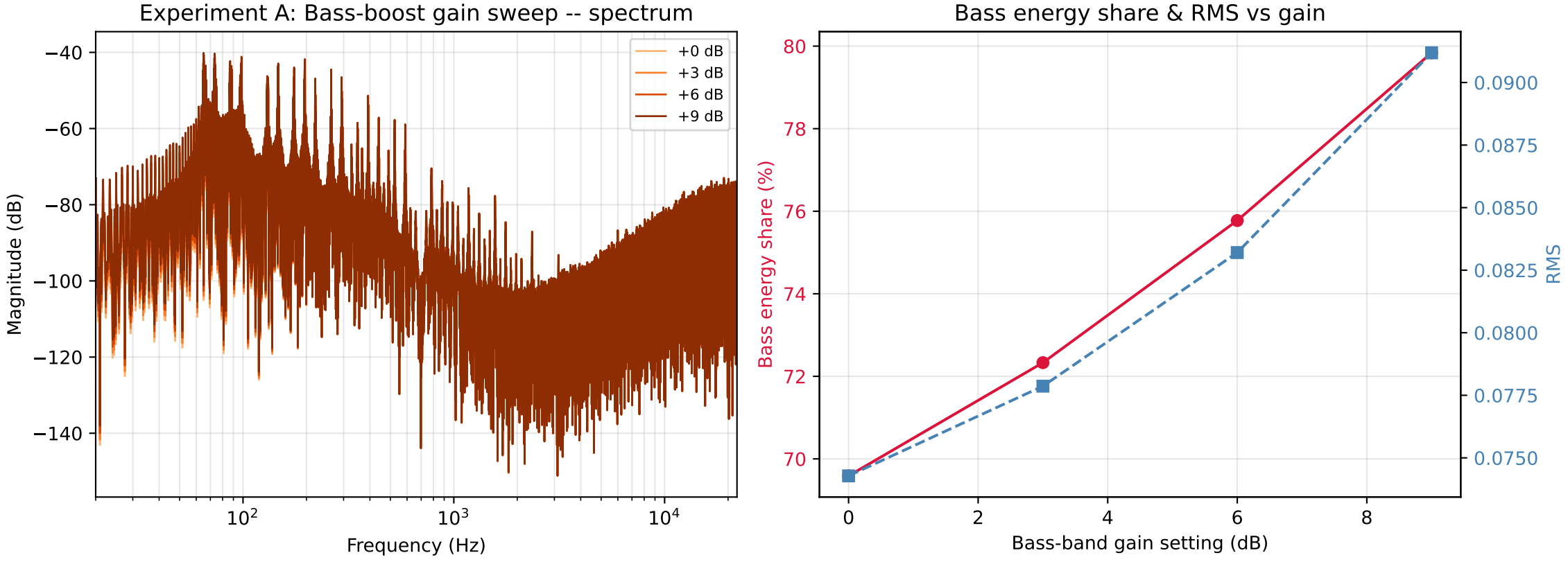
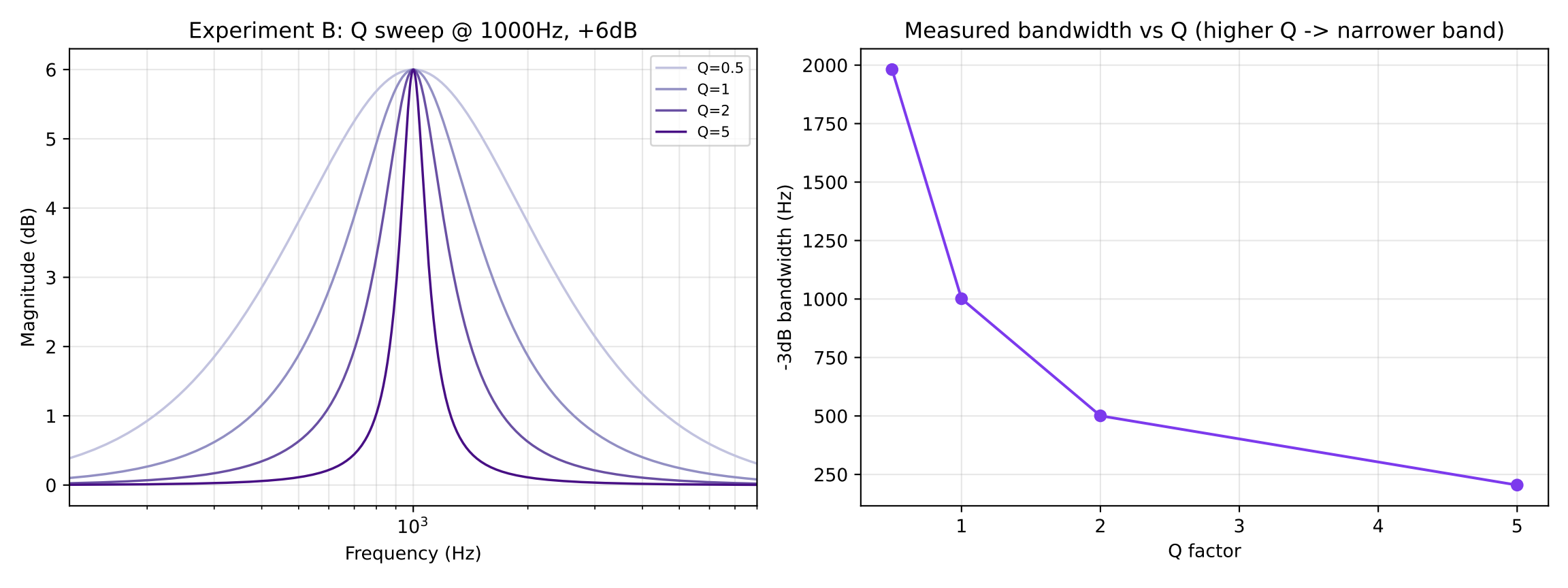
## Experiment A: Bass Boost Gain Sweep

| Gain (dB) | Bass energy % | RMS | Peak |
|---|---|---|---|
| +0 | 69.6% | 0.0743 | 0.9000 |
| +3 | 72.3% | 0.0779 | 0.9019 |
| +6 | 75.8% | 0.0832 | 0.9091 |
| +9 | 79.8% | 0.0912 | 0.9193 |


Experiment A: Bass-boost gain sweep -- spectrum
Magnitude (dB)
−40
−60
−80
−100
−120
−140
$10^2$
$10^3$
$10^4$
Frequency (Hz)
+0 dB
+3 dB
+6 dB
+9 dB
Bass energy share & RMS vs gain
Bass energy share (%)
80
78
76
74
72
70
RMS
0.0900
0.0875
0.0850
0.0825
0.0800
0.0775
0.0750
0
2
4
6
8
Bass-band gain setting (dB)

## Experiment B: Q Factor Sweep

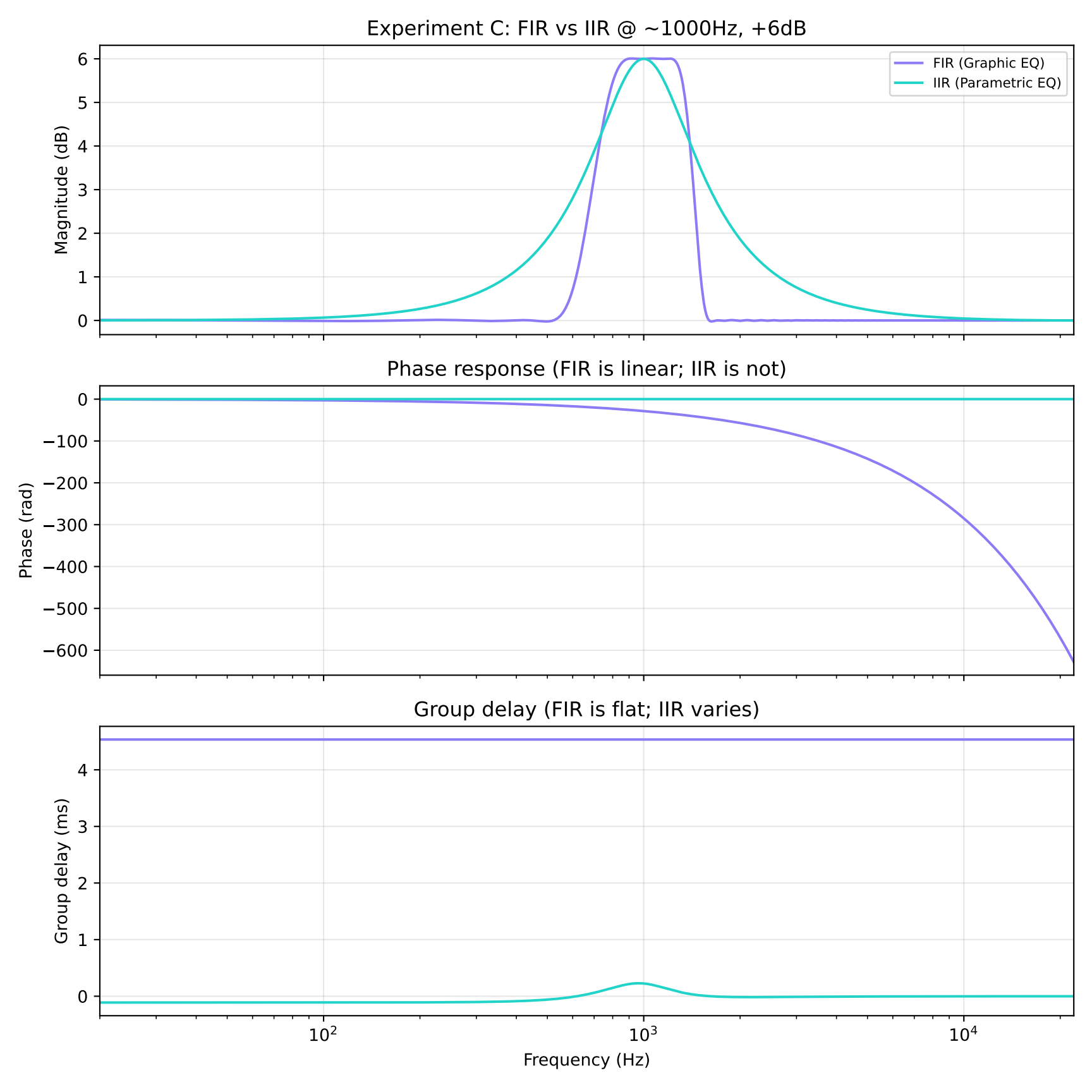
| Q | Measured gain (dB) | -3dB Bandwidth (Hz) |
|---|---|---|
| 0.5 | +6.00 | 1981.1 |
| 1 | +6.00 | 1001.3 |
| 2 | +6.00 | 500.6 |
| 5 | +6.00 | 204.6 |


Experiment B: Q sweep @ 1000Hz, +6dB
Q=0.5
Q=1
Q=2
Q=5
Magnitude (dB)
Frequency (Hz)
$10^3$
Measured bandwidth vs Q (higher Q -> narrower band)
-3dB bandwidth (Hz)
Q factor

## Experiment C: FIR vs IIR

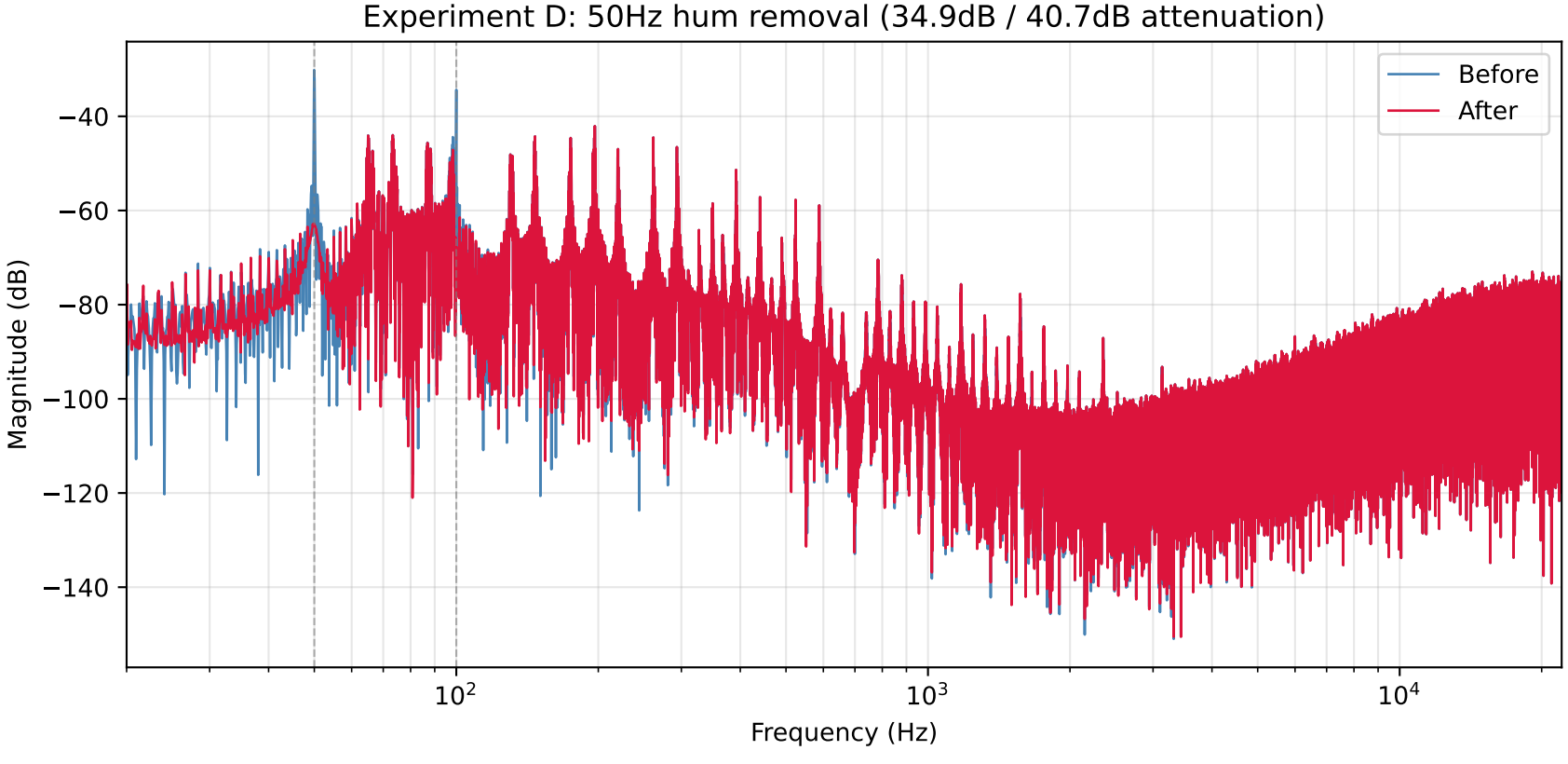
```
Target: 1000Hz, +6dB (graphic band used: 1000Hz)

FIR -- coefficients: 401, processing time: 408.25 ms
IIR -- coefficients: 6, processing time: 3.58 ms
```


Experiment C: FIR vs IIR @ ~1000Hz, +6dB
FIR (Graphic EQ)
IIR (Parametric EQ)
Magnitude (dB)
6
5
4
3
2
1
0
Phase response (FIR is linear; IIR is not)
Phase (rad)
0
−100
−200
−300
−400
−500
−600
Group delay (FIR is flat; IIR varies)
Group delay (ms)
4
3
2
1
0
$10^2$
$10^3$
$10^4$
Frequency (Hz)

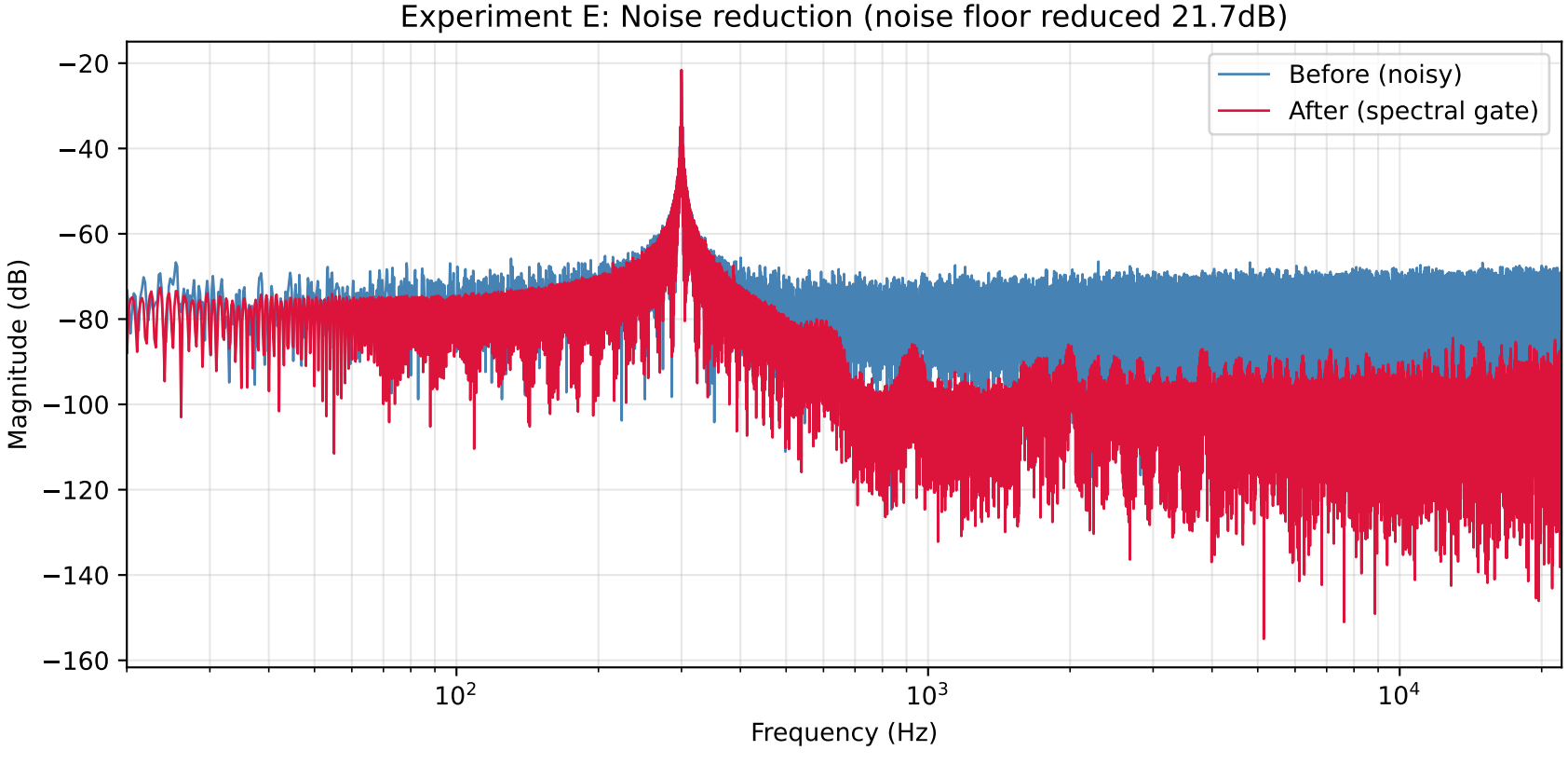
## Experiment D: 50Hz Hum Removal

```
Fundamental (50Hz) attenuation : 34.9 dB
2nd harmonic (100Hz) attenuation: 40.7 dB
```


Experiment D: 50Hz hum removal (34.9dB / 40.7dB attenuation)
Before
After
Magnitude (dB)
−40
−60
−80
−100
−120
−140
$10^2$
$10^3$
$10^4$
Frequency (Hz)

## Experiment E: Noise Reduction

```
Noise-only RMS before: 0.0499
Noise-only RMS after : 0.0041
Noise floor reduction: 21.7 dB
```


Experiment E: Noise reduction (noise floor reduced 21.7dB)
Before (noisy)
After (spectral gate)
Magnitude (dB)
−20
−40
−60
−80
−100
−120
−140
−160
$10^2$
$10^3$
$10^4$
Frequency (Hz)